

R.O.B. VISION

VIRTUAL ROBOT COMPANION / MK.01

TECHNICAL REFERENCE

Architecture, frame protocol, virtual model, pairing, and verification

VERSION 0.1.7

UNO Q · RETROPIE · BATOCERA



R.O.B. Vision technical reference

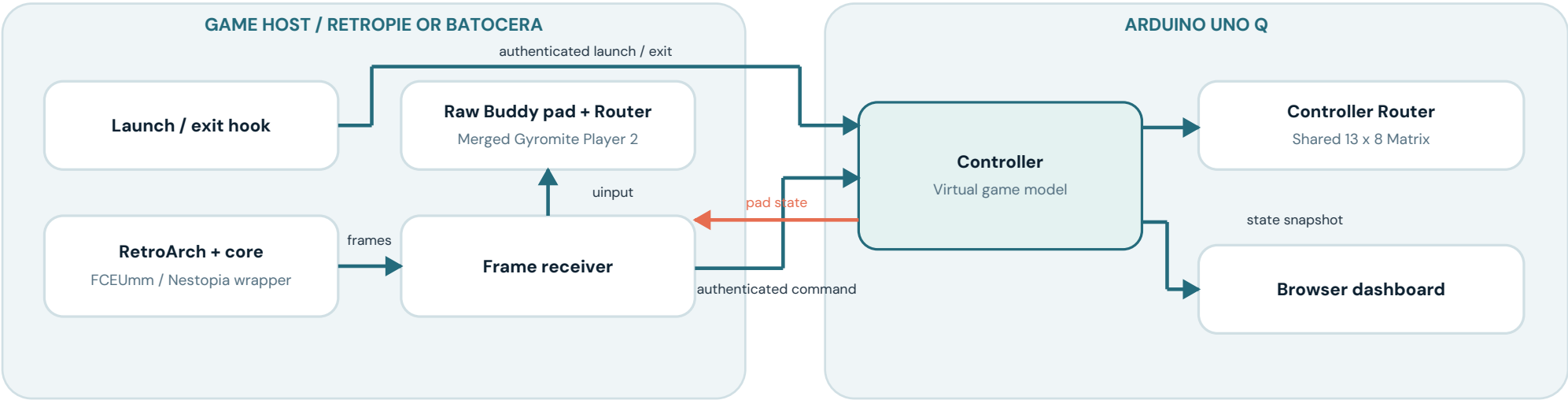
This is the implementation reference for the current system. The Arduino UNO Q owns Buddy's virtual pose and pieces. RetroPie, Batocera, or Recalbox runs Gyromite or Stack-Up. A browser on a laptop, tablet, or phone renders the controller's state. Automatic commands come from emulator-rendered NES frames; the current application has no camera input path.

Find an interface

Investigation	Section
Game selection or stale session	Game identity and session lifecycle
Command decoding or retry	Frame classification
Piece alignment or collision	Virtual model
Gyromite gate input	Gyromite return path
Wrong RetroArch player	Session routing
Pairing or API access	Pairing and trust boundary
Failure diagnosis	Recovery and verification

The following sections distinguish the observed signal, the accepted model action, and the emulator's returned input. Each is a separate verification boundary. For a player walkthrough, use the [User Guide](#).

System and responsibilities



Component	Responsibility	Source
Game registry	Exact supported ROM basenames and game IDs	config/games.json, tools/identify_game.py
Console launch hook	Start/end events; never robot motion	RetroPie runcommand hooks, Batocera zz-robvision-game, or Recalbox process monitoring; tools/notify_game.py
Core wrapper	Classify each emulated NES video frame and pass the original video onward	deploy/retropie/rob_vision_fceumm_proxy.c
Console receiver	Check sender, decode commands, report heartbeats, drive virtual Player 2	tools/retropie_frame_hook.py, tools/retropie_controller2.py
UNO Q controller	Game selection, virtual model, pairing, snapshot API	controller/service.py, controller/model.py, controller/pairings.py
UNO Q presentation	Browser views on 8101; named product display requests	python/main.py, controller/matrix.py, matrix/manifest.json, dashboard/
Shared UNO Q Router	Entry page on 80, live-session selection, boot-bound input leases, sole Matrix sketch	controller_router_portal/host/concurrent.py, host/display_runtime.py, app/sketch/sketch.ino

The UNO Q is the source of truth for game, pose, pieces, and buttons. The browser polls and animates that state; it does not decode game frames. A directly opened `file:///dashboard` is an independent scripted preview, not the live controller.

Game identity and session lifecycle

The registry matches exact, case-insensitive Gyromite (World) and Stack-Up (World) basenames with `.nes`, `.zip`, or `.7z` extensions. The resolver accepts NES/Famicom metadata, but the installed frame integration and ROM directories are for **NES**. A renamed archive needs a registry entry. The launched archive name matters, not the name inside it. No ROMs or extracted game art ship with the project.

1. A console hook posts a credentialed `start` event with the system and filename to `/api/launch`. The UNO Q resolves it and initializes the matching game's virtual pieces. An unknown title clears the game.
2. The selected R.O.B. Vision wrapper delegates libretro calls to the installed original FCEUmm or Nestopia core and observes video frames. A plain core can play the game but does not provide the frame link.
3. The receiver accepts frames only from an approved wrapper running a registered ROM. It decodes a complete command before posting it to the UNO Q. A launch event alone never moves Buddy.
4. The receiver polls the UNO Q and drives its virtual Controller 2 only while Gyromite is the active RetroArch process. Stack-Up has no controller-return buttons.
5. An end event clears the selected game. The receiver releases the pad. Its RetroArch scan can replay a recognized launch after a UNO Q restart, with retries throttled to two seconds.

The latest authenticated console launch selects the active game and console. While that ownership remains, a different console's frame command is rejected. A manual browser game selection clears console ownership and remains unowned until another authenticated launch. Removing the active console's pairing clears its session. When idle, an online receiver label may refer to either paired console; it is not evidence of a running game.

Frame classification and command protocol

The wrapper intercepts the libretro video callback, classifies the frame, and then forwards the video to RetroArch. It samples a 5-by-5 grid plus nine central picture points. Near-black pixels count as dark; a green-dominant pixel counts as light. A frame is 0 with at least 23 dark grid points, 1 with at least 23 green grid points or eight green central points, and N otherwise. The central sample handles dark borders on Test artwork. These are **emulated frames**, so monitor refresh and camera frame rate are outside the input path.

Each nonblocking Unix datagram is five bytes: a native 32-bit frame index and ASCII 0, 1, or N. It goes to `/run/rob-vision/frames.sock`. The receiver uses Unix process credentials, RetroArch's command line, an approved wrapper path, and exact ROM identity to authenticate the local sender; it rechecks the process at least every half second. A new sender or game resets the decoder.

Both games encode one primitive per 13-frame word, `000101w1x1y1z`, with four variable bits. Stack-Up schedules an extra dark frame after the data word. The [ROM analysis](#) has the examined ROM hashes, CPU addresses, palette proof, and reproducible analysis command.

The inspected Gyromite image has SHA-256 `bf1b323ba39c84b964f93127598b267ff3f16cbf99c533ae1e9b8332232b3e5b`; Stack-Up has `3ab6bc99246783a6bf7083481027fe358ccdb147dcf1165eaf08aaf4e1b06548`. Both are iNES mapper-0 images with 32 KiB program and 8 KiB character ROM. Gyromite's palette-choice table at CPU `$A6FC` supplies 13 entries per populated command slot; its routine at `$A679-$A697` reads them in reverse order. Stack-Up's sender at `$B1DE-$B237` uses the template at `$B24A`. Palette lookups tie the extracted bits to black and green screen frames. These addresses and hashes identify the examined ROM variants, not every regional or patched copy.

Primitive	Gyromite word	Stack-Up word	Model effect
Open	0001011101110	Same	Open hands; release a held piece if placement is valid
Close	0001010111110	Same	Close hands; possibly pick up a piece or block segment
Left	0001010111010	Same	Turn one station left
Right	0001011101010	Same	Turn one station right
Up	0001010111011	0001011111010	Gyromite: two levels; Stack-Up: one
Down	0001011111011	0001010101110	Gyromite: two levels; Stack-Up: one
Ready light	0001011101011	Same	Brief status light; no movement

`ExactFrameDecoder` requires consecutive frame indices and a complete word allowed for the selected game. A gap, N in the candidate, wrong Up/Down code, or incomplete word does nothing. Separate identical complete words are valid. The frame hook buffers up to 32 decoded commands before the receiver drains them into its delivery queue. The receiver retains a command for up to 12 seconds and retries failed HTTP delivery every 200 ms, preserving the first command while Router selects the game’s controller. It waits while process scanning has not yet identified the game, and drops commands when their wrapper process exits, the game changes, the active console differs, or the retention limit expires. The UNO Q deduplicates retained retries by sender PID and ending frame index, keeping the last 128 keys. It also rechecks game, pattern, PID, index, and active console. This prevents an ordinary retry from repeating motion; it is not a durable exactly-once guarantee across restarts.

Test detection is separate from command decoding. At least 36 consecutive green frames or 12 alternating edges mark a Test field. A recent authenticated heartbeat pulses the Test light. A complete ready-light word lights it steadily for at most one second. Neither changes pose or pieces. The signal does not reveal the game’s menu state through memory.

Virtual model

`GET /api/state` returns schema 1 with `game`, `robot`, `events`, `input`, `test`, and `link`. The UNO Q keeps a sequence number and the latest 30 display events, not a durable event log. Browsers poll every 500 ms. A refresh reads the current snapshot. A fresh `input . frame_hook` means matching game frames recently reached the receiver; it does not prove that the game gate moved or a Stack-Up objective was completed.

Gyromite

The model tracks two gyros, six height levels, grip, held gyro, spin deadlines, and manual Gate Assist deadlines across five ordered stations: holder B, holder A, red pad, blue pad, spinner. Home is level 6. Gyromite Up/Down moves two levels, so game commands reach levels 6, 4, and 2; all five accessories meet the hands at level 2 in the virtual model. The original manual’s positions 1–5 are horizontal base slots, not vertical levels. Their different screen heights are perspective. A carried gyro must rise to at least level 4 before turning; level 6 remains available for greater clearance. This is a virtual collision rule rather than a numbered rule from the original manual. Stack-Up Up/Down moves one level and uses all six positions. Placing a gyro on the spinner starts an illustrative 55-second spin clock. A spinning gyro resting on a colored pad presses its button. A held gyro at a colored pad at level 2 may press it without spinning; lifting it releases the button. The controller does not measure physical spin or inspect gate pixels.

Fast Gates is a manual browser control. It temporarily puts an available gyro on a pad and holds that button for at most 60 seconds, then returns the gyro to its holder on release or timeout. Game-frame movement is rejected while Gate Assist is held. Red and blue are independent. Selecting a game or Home rebuilds the pieces; Emergency Stop selects no game.

Stack-Up

The model tracks five trays, five distinct blocks, six height levels, one station, and one grip. It starts all blocks on Tray 3, bottom to top green, yellow, blue, white, red. Left/Right moves one tray; Up/Down moves one level. Closing at a block height lifts that block and everything above it as an ordered segment. A carried segment must clear another tray; release requires the next free level. Invalid moves leave pieces in place, and a conservation check rejects loss or duplication.

Direct, Memory, and Bingo transmit the same six movement primitives. The controller does not receive target pattern, score, or victory state. It currently uses one starting arrangement; distinct mode-specific historical layouts and extended Memory/Bingo play are outside the verified model claim.

Gyromite return path

The console receiver creates a Linux `/dev/uinput` pad named `R.O.B. Vision Controller 2`. It polls the UNO Q every 50 ms with a 250 ms request timeout, validates schema 1, and emits changed red/blue states with a sync event. It releases both when Gyromite is not the active RetroArch process, the paired source is not selected, or no good response arrives for 750 ms. Exit and service shutdown also release them.

Runtime setting	Current value	Owner
Browser snapshot poll	500 ms	dashboard/live.js
Receiver poll / HTTP timeout / stale pad release	50 / 250 / 750 ms	tools/retropie_controller2.py
RetroArch process scan / launch replay throttle	250 ms / 2 s	tools/retropie_controller2.py
Frame-link freshness / receiver-online freshness	1 s / 3 s	controller/service.py
Recent event history / retry keys	30 events / 128 keys	controller/service.py
Gyro spin / manual gate maximum	55 / 60 s	controller/model.py
Game registry	Installed config/games.json on each console	Console receiver; UNO Q trusts its authenticated game assertion
Console registry and Router editor	HTTP port 8769	Paired console, proxied through UNO Q Setup
Frame socket	/run/rob-vision/frames.sock	Console receiver

Controller Router is packaged with R.O.B. Vision. Its config schema is shared with VirtualGlove: existing Player 1-4 assignments are preserved, and Buddy is added as a Player 2 source. On a fresh console, configured EmulationStation controllers seed the corresponding player slots. The Setup Router editor can reassign physical sources and tests button activity. The raw Buddy pad has a fixed two-button Linux mapping; the receiver swaps red and blue before Router translates them into canonical merged A/B buttons. The receiver and wrapped-game launch hooks never write RetroArch port assignments.

The routing engine is now the vendored `router_shared` package from the separate Controller Router library, with identical source in R.O.B. Vision and VirtualGlove. It owns controller discovery, stable assignments, uinput outputs, and RetroArch indexes. R.O.B. Vision consumes Router pairing, retains its console API, and performs its first-pair Buddy assignment in `tools/controller_router_setup.py`. Buddy's two-button identity and mapping live in `config/router_sources.json`, which the generic library reads as a virtual-source descriptor. VirtualGlove keeps its signed console-request adapter and its gesture-player choice. A source change is made in the library and synchronized to both projects; neither project edits its vendored engine independently.

Session routing across RetroArch versions

Controller Router resolves its merged pads by unique name and vendor/product identity before RetroArch executes. RetroPie 1.19.1 uses a temporary appended configuration with current udev indexes. Supported newer executables use strict native reservations; unknown builds use the legacy path. Native mode seeds a complete initial index permutation to avoid the RetroArch 1.20 reservation allocator's duplicate-index crash.

Runtime routing does not edit saved RetroArch configuration. Installers register the shared adapter in RetroPie's `emulators.cfg`, retaining `pi:pi` ownership and existing environment, arguments, native hand input, and cabinet hooks. Batocera uses a narrow Libreto generator overlay that inserts the adapter after config generation. Existing appended settings remain ahead of the temporary routing file. Automatic configuration save on exit is disabled for routed sessions so transient indexes cannot become saved settings.

Physical sources reconnect to their saved players without destroying merged outputs. Changing assignments applies on the next launch. If Router fails during play, the adapter reports the loss and Router refuses to rebuild outputs until the game ends; recovery requires relaunch. Logs include mode, name, VID/PID, event node, and launch slot. See Controller Router's `ROUTING_VALIDATION.md` for the dated tests and their scope.

Pairing, API, and trust boundary

Controller Router serves the app entry page on port 80. R.O.B. Vision's Linux service serves its browser on port 8101 and console API on port 8766. The UNO Q's existing hostname and mDNS service provide its `.local` address; R.O.B. Vision does not advertise a separate name. A reachable LAN IP works too. Wi-Fi and Ethernet carry the same HTTP protocol. Keep ports 80 and 8766 on a trusted LAN. For console pairing and registry requests, an App Lab resolver sidecar forwards bounded `.local` lookups to the UNO Q host's Avahi socket. The controller checks that the resolved address is private before connecting. This is needed because the App Lab container does not inherit the host's mDNS name resolution.

Endpoint	Effect	Access
GET /api/state	Browser snapshot; receiver poll carries frame/Test headers	LAN read; credential verified when supplied
GET /api/matrix/state	Matrix bridge status	LAN read
POST /api/launch	Console start/end and registered game assertion	Paired console credential
POST /api/games	Read, validate, save, or restore one console registry	Trusted-LAN browser; UNO Q proxies with that console's private credential
POST /api/router	Read, test, save, or restore that console's controller assignments	Trusted-LAN browser; UNO Q proxies with that console's private credential
POST /api/emulator/command	One ROM-derived word	Paired credential and active-source checks

Endpoint	Effect	Access
POST /api/game, /api/command, /api/gate-assist	Manual browser controls	Trusted-LAN browser, JSON and same-origin checks
POST /api/router-pairing	Import/revoke app credentials	Private host capability; never browser access
POST /api/matrix/pairing	Show or clear pairing cue	Trusted-LAN browser

Shared device pairing

Controller Router is the connection authority. The UNO Q host broker runs secure Setup on TCP **8444** and stores schema-1 connections under `/home/arduino/.local/state/controller-router/connections.json`. The persistent console TLS service listens on TCP **55359**. Router has a management credential; VirtualGlove and R.O.B. Vision have distinct, independently revocable credentials. Browser responses contain only public connection and readiness fields.

`router_shared.pairing.Peer` checks the console certificate before sending a code or credential. The CR1 code contains a 100-bit certificate fingerprint prefix and a 60-bit authorization value. Console windows last 300 seconds, accept one successful transaction, and lock after five incorrect codes. Physical Matrix confirmation lasts 120 seconds and allows five attempts. Requests are bounded; servers allow at most 16 concurrent workers and require TLS 1.2 or later. Secure browser writes require matching HTTPS Origin, a fixed action header, and JSON content.

Provisioning uses prepare, commit, and finalize. Private journals retain the previous Router registry and app configuration; interrupted or failed transactions restore those snapshots. Successful changes keep private before-connection backups. Legacy migration verifies an HMAC over a fresh nonce, canonical console ID, and observed TLS certificate; adoption signs the new connection transcript. Hostnames alone never authorize consolidation. Conflicting records remain unchanged for an explicit re-pairing choice.

Products expose `/api/router-pairing` only to a capability-bearing local host request. Capability files are named `data/router-pairing-adapter-token` and must not be exposed to browsers. Adapters import app credentials into live caches without replacing ROM registries, calibration, or player settings. Router polls for newly installed console and UNO adapters and provisions them using its pinned management connection. Disabled app access stays disabled.

The console helper has `/identity`, `/pair`, `/legacy-proof`, `/adopt`, `/manage`, and `/router` interfaces. `/manage` uses Router's credential for inspection, provisioning, and removal. `/router` uses the same credential with RouterStore revision checks for assignments and system policy. It never writes saved RetroArch configurations. Buddy's first-pair adapter adds only its Player 2 source to Router configuration; emulator configuration migration remains installation-only.

Back up the UNO Q connection registry, its `tls` directory, and the private before-connection backups. On RetroPie, also back up `/var/lib/controller-router/link/{console-id, adapters.json, connection.json, certificate.pem, private-key.pem}` and each product's credential files. Restore requires the separately installed shared link service. Keep backups private; never include credentials in diagnostics or support reports. Do not snapshot a provisioning transaction in progress.

Matrix and browser

Controller Router owns the 13×8 framebuffer through its App Lab Matrix service. R.O.B. Vision supplies `matrix/manifest.json`; `controller/matrix.py` translates model state into named animation requests over the local Unix socket. Priority within the product is pairing, Test/ready, selected game, then clear. Games use `gyromite` and `stack_up`; Test uses `test` or `test_flash`; pairing uses the product's `pairing` animation. Accepted movements request `hint_left`, `hint_right`, `hint_up`, `hint_down`, `hint_open`, or `hint_close`. Idle clears the product request, leaving Router's neutral artwork. Only the selected product's requests are accepted. R.O.B. Vision refreshes its request at least once per second. Router expires ordinary display requests after five seconds without refresh; the shared sketch returns to neutral when no frame arrives for 1.8 seconds. The separate legacy R.O.B. Vision sketch remains a standalone development fallback and is not flashed by the normal installer. The [Matrix Display Guide](#) explains the player-facing cues.

Mission animates the model, accessory views, Game Table, Pose Preview, vitals, and recent activity from the same snapshot. Setup handles pairing, link checks, Test status, and manual game checks. A paired console may be online while no game is selected. Browser demo runs while paired and idle; a live game takes authority. The UI cannot establish Hector's location, Stack-Up scoring, or full-game completion.

Installation and platform integration

The versioned release installer verifies its source package before extraction and dispatches to UNO Q or console installation. UNO Q replacement preserves pairing data and registers the product with the shared Router. A newer compatible Router remains installed while the new product is still registered.

On the UNO Q:

1. Controller Router owns the sole App Lab sketch and Matrix service. Product Linux services run continuously in separate Compose projects.
2. Port 80 serves app selection, 8100 serves VirtualGlove, and 8101 serves R.O.B. Vision. The chooser opens running sites; it does not start or stop products.
3. Boot begins without a selected product. Authenticated product session state selects the game's controller; a browser is not required.
4. Router grants one boot-bound lease valid for two seconds. The heartbeat sleeps 250 ms between reconciliations; network checks can lengthen the interval.
5. Game exit revokes a game-owned selection. Manual switching is blocked during play. Conflicting live sessions revoke both leases.
6. A missing or expired lease denies product input. Receiver watchdogs release held controls.

The installer prints `.local` and LAN IPv4 links. The product **Apps** link is shown only when both controller products are installed.

RetroPie installs launch/end hooks, a systemd receiver, a virtual-pad RetroArch profile, and FCEUmm/Nestopia wrapper launch choices. Its installer builds wrappers against installed cores and installs the shared Controller Router. First pairing assigns Buddy to Player 2. It retains an existing per-ROM supported core choice rather than forcing every host to FCEUmm. EmulationStation must be closed while the virtual input service is replaced.

Batocera selects a packaged wrapper for x86_64, 32-bit x86, AArch64, ARMv7, ARMv6, or RISC-V 64, or compiles one locally for an unmatched ABI when a C compiler is available. It then installs a separate ROBvision service and game hook, runtime core overlay, and persistent per-ROM overrides. The installer preserves an existing supported Nestopia choice; otherwise it selects an available FCEUmm or Nestopia wrapper for registered ROMs. The installer checks for the required NES core, libretro info, and loadable native wrapper before writes. It suspends and resumes EmulationStation around receiver restart. When VirtualGlove is installed and enabled, ROBvision waits for its core overlay before adding its own wrappers; it does not stop or disable VirtualGlove.

Recalbox 10.x installs its receiver and frame wrappers in the persistent share. At boot, a narrow overlay exposes the FCEUmm and Nestopia wrappers and registers them in the NES system list. Exact registered ROM filenames receive `.recalbox.conf` sidecars, preserving unrelated ROM choices. Controller Router mounts a patched Libretro generator for the current boot and adds temporary, identity-based routing settings only for enabled systems. The receiver identifies the active game process and reports start and end to the UNO Q. The pairing helper uses Recalbox's

Avahi socket when NSS . local lookup is unavailable. Recalbox 10.1.1 on the `rpizero2` target passed bounded Gyromite and Stack-Up launches, paired UNO Q game selection, and reboot recovery. A direct Gyromite receiver trace confirmed the wrapper delivered a neutral frame packet from the identified RetroArch process; movement from a human-played command remains to be verified on this host. The [installation guide](#) has the supported command and prerequisites; the [configuration reference](#) lists tunable defaults.

Recovery and verification boundary

Condition	Behavior
Browser refresh	Reads UNO Q snapshot without resetting
Unknown title or game exit	Clears selected game; virtual buttons release
UNO Q restart during a registered game	Receiver can replay launch; frame link resumes after authenticated polling and wrapper frames
Missing, non-light, or malformed frame	Drops partial word; no repeated motion
No good receiver response for 750 ms	Both Gyromite buttons release
No authenticated receiver poll for three seconds	Console status becomes offline
No matching game frames for one second	<code>input.frame_hook</code> becomes false
Router-managed lease missing, invalid, expired, or from an earlier boot	Product input is denied; no standalone fallback
No authenticated receiver check-in for ten seconds	R.O.B. Vision clears its live session
No Matrix frame for 1.8 seconds	Shared sketch resumes Router's neutral animation

Operational checks

Observation	Interpretation	Next check
Game selected, frames waiting	Launch hook reached UNO Q; the approved wrapper has not produced fresh matching frames	Confirm the registered ROM launched with the R.O.B. Vision FCEUmm or Nestopia choice
Game frames linked, no movement	Fresh frames arrived; no valid command may have completed, or the model may have blocked one	Send a Direct-mode command and read the latest decoded, <code>action</code> , or blocked event
Console online, game idle	Receiver polling works; no recognized active game	Check exact ROM basename and console launch hook
Gyromite action visible in Mission, gate unchanged	Model changed, but the return path or game state may be wrong	Check Player 2 mapping, active RetroArch process, and independent blue/red holds
Test light pulses, pose unchanged	Test field recognized as status	Leave Test mode and confirm the light clears before a Direct command
Matrix stays neutral while a game is selected	Product may lack the lease or its Matrix requests may not be delivered	Check Router <code>/api/state</code> , the product <code>/api/matrix/state</code> , and the shared Matrix service

The checks support different claims: a recognized ROM proves selection; a fresh frame link proves source delivery; a decoded event proves a complete word; an `action` event proves the virtual model accepted it; a visible gate response proves Controller 2 returned input to Gyromite. Do not collapse these into one "connected" result.

Verified on supported test hosts	Still outside the demonstrated claim
Real launch and exit detection for both games	Unknown-title interactive launch on every host
All six Direct-mode movement commands from both games under FCEUmm	Long unattended sessions and measured missed-command rate
Movement commands under Nestopia for both games	Full six-command Nestopia sessions under every game mode
Gyromite blue-gate response and return under both cores	Full-game gate automation and game outcomes
Batocera 43.1 x86_64 reboot with VirtualGlove coexistence	Other Batocera board/version combinations
Automated model, decoder, installer, and pairing tests	Stack-Up Memory/Bingo behavior and complete matrix visual checks

A valid frame link alone does not imply model success: an accepted command can be blocked at a boundary. The browser does not know Hector's position or the Stack-Up goal. The matrix's active modes are implemented, while complete direct visual checks remain open.

The [verification plan](#) tracks open checks. The [FCEUmm report](#) and [Nestopia report](#) preserve dated evidence. The [engineering journey](#) explains how camera testing led to the frame link. Earlier [camera row tests](#) and [optical field notes](#) are historical research and do not describe the current input path.